

# linearcoefficients – version 1.0.0

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## Abstract

This package helps dealing with the notation for linear elastic/piezoelectric materials by defining commands for fields and coefficients as well as support for easy switching between different notations. There is also support for handling the superscripts denoting the conditions for the coefficients.

## 1 Notation

This package provides commands for the symbols of the physical quantities of a piezoelectric as well as for the linear coefficients relating them. The actual symbols are defined according to specified a notation. The primary objective of this functionality is to add a layer of abstraction that make reuse of L<sup>A</sup>T<sub>E</sub>X code in a context that requires different notation.

Some notations are predefined: One mostly congruent with the definitions in the IEEE standard on piezoelectricity [?]. It is set with the command `\IEEENotation` or by passing the option `IEEE` to the package. The other notations follow Nye [?], Ikeda [?] and Mason [?], The corresponding options are `Nye`, `Ikeda` and `Mason` with respective commands `\NyeNotation`, `\IkedaNotation`, `\MasonNotation`.

Any number of notational options can be given to the package. The last one in the options list is the one chosen. If none is given, the package defaults to the IEEE-like notation. This defines the notation globally in the document. The predefined notations are given in table ?? which shows the physical quantity, its corresponding symbol in each predefined notation and the command to generate it.

The definitions can be changed within the local scope (group). For example, in a document where the globally defined notation is the IEEE-like notation, the code

```
\[
  {\NyeNotation\stress_{ik} = \stiffness_{ijkl}\strain_{kl}}
  \qquad\mbox{vs.}\qquad
  {\noelectrical \stress_{ik} = \stiffness_{ijkl}\strain_{kl}}.
\]
```

produces

$$\sigma_{ik} = c_{ijkl}\epsilon_{kl} \quad \text{vs.} \quad T_{ik} = c_{ijkl}S_{kl}.$$

Table 1: Symbols for the predefined notations (sans indices).

| Quantity                         | Command                          | IEEE       | Nye        | Ikeda      | Mason      |
|----------------------------------|----------------------------------|------------|------------|------------|------------|
| electric field                   | <code>\efield</code>             | $E$        | $E$        | $E$        | $E$        |
| electric polarization            | <code>\polarization</code>       | $P$        | $P$        | $P$        | $P$        |
| electric flux density            | <code>\dfield</code>             | $D$        | $D$        | $D$        | $D$        |
| stress                           | <code>\stress</code>             | $T$        | $\sigma$   | $\sigma$   | $T$        |
| strain                           | <code>\strain</code>             | $S$        | $\epsilon$ | $\epsilon$ | $S$        |
| temperature                      | <code>\temperature</code>        | $\theta$   | $T$        | $T$        | $T$        |
| entropy density                  | <code>\entropydensity</code>     | $\sigma$   | $s$        | $s$        | $S$        |
| stiffness coefficient            | <code>\stiffness</code>          | $c$        | $c$        | $c$        | $c$        |
| compliance coefficient           | <code>\compliance</code>         | $s$        | $s$        | $s$        | $s$        |
| permittivity                     | <code>\permittivity</code>       | $\epsilon$ | $\kappa$   | $\epsilon$ | $\epsilon$ |
| impermittivity                   | <code>\impermittivity</code>     | $\beta$    | $\beta$    | $\beta$    | $\beta$    |
| piezoelectric cofefficient       | <code>\piezoh</code>             | $h$        | $h$        | $h$        | $h$        |
| piezoelectric cofefficient       | <code>\piezog</code>             | $g$        | $g$        | $g$        | $g$        |
| piezoelectric cofefficient       | <code>\piezoe</code>             | $e$        | $e$        | $e$        | $e$        |
| piezoelectric cofefficient       | <code>\piezod</code>             | $d$        | $d$        | $d$        | $d$        |
| heat capacity                    | <code>\heatcapacity</code>       | $?$        | $C$        | $C$        | $C$        |
| pyroelectric coefficient         | <code>\pyroelec</code>           | $?$        | $p$        | $p$        | $q?$       |
| heat of polarization             | <code>\heatofpolarization</code> | $?$        | $t$        | $t$        | $?$        |
| coefficient of thermal expansion | <code>\thermalexpansion</code>   | $?$        | $\alpha$   | $\alpha$   | $\alpha$   |
| heat of deformation              | <code>\heatofdeformation</code>  | $?$        | $f$        | $f$        | $?$        |

## 2 Free variables

It is customary to denote the conditions under which a linear coefficient is measured by superscripts. For example, a stiffness constant measured at constant electric field and temperature is written  $c_{ij}^{E,\theta}$  in the IEEE-like notation. It is possible to accomplish this by writing `c^{\efield,\temperature}_{ij}`, but the package provides notation to automate the inclusion of such superscripts.

The automated notation hinges on declaration of at most one field variable from each of the domains mechanical, electrical and thermal as free variables. The package then includes the relevant variables among those that are declared as independent in the supercripts.

The command `\freestress` declares stress as a free variable in the mechanical domain. Since the conjugate variable strain then must be dependent, any prior declaration of it as free is nulled by the command. Likewise `\freestrain` declares strain as the free variable in the mechanical domain. In the electrical domain the similar commands are `\freeefield` and `\freedfield` for respectively the electrical field and the electric flux density. In the thermal domain the commands are `\freetemperature` and `\freeentropydensity` for respectively temperature and entropy density.

The commands `\nomechanical`, `\noelectrical` and `\nothermal` clear all declarations of free variables in the domain indicated by the name of the command. The command `\nofree` clears all declarations of free variables.

As an example usage, consider the code

Table 2: Examples of IEEE-like notation for coefficients depending on which fields are declared free.

| Free           | Coefficients                                                                                                                                                |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| –              | $c_{ijkl}, s_{ijkl}, \epsilon_{ij}, \beta_{ij}, h_{ikl}, g_{ikl}, e_{ikl}, d_{ikl}$                                                                         |
| $E$            | $c_{ijkl}^E, s_{ijkl}^E, \epsilon_{ij}, \beta_{ij}, h_{ikl}, g_{ikl}, e_{ikl}, d_{ikl}$                                                                     |
| $D$            | $c_{ijkl}^D, s_{ijkl}^D, \epsilon_{ij}, \beta_{ij}, h_{ikl}, g_{ikl}, e_{ikl}, d_{ikl}$                                                                     |
| $E, T$         | $c_{ijkl}^E, s_{ijkl}^E, \epsilon_{ij}^T, \beta_{ij}^T, h_{ikl}, g_{ikl}, e_{ikl}, d_{ikl}$                                                                 |
| $E, S$         | $c_{ijkl}^E, s_{ijkl}^E, \epsilon_{ij}^S, \beta_{ij}^S, h_{ikl}, g_{ikl}, e_{ikl}, d_{ikl}$                                                                 |
| $E, S, \theta$ | $c_{ijkl}^{E,\theta}, s_{ijkl}^{E,\theta}, \epsilon_{ij}^{S,\theta}, \beta_{ij}^{S,\theta}, h_{ikl}^\theta, g_{ikl}^\theta, e_{ikl}^\theta, d_{ikl}^\theta$ |
| $E, S, \sigma$ | $c_{ijkl}^{E,\sigma}, s_{ijkl}^{E,\sigma}, \epsilon_{ij}^{S,\sigma}, \beta_{ij}^{S,\sigma}, h_{ikl}^\sigma, g_{ikl}^\sigma, e_{ikl}^\sigma, d_{ikl}^\sigma$ |

```

{\NyeNotation
\begin{align*}
\backslash strain_{ij} &= \backslash compliance_{ijkl} \backslash stress_{kl} + \\
&\quad \backslash piezod_{kij} \backslash efield_k \\
&\quad + \backslash thermalexpansion_{ij} \backslash temperature \\
\backslash dfield_i &= \backslash piezod_{ikl} \backslash stress_{kl} \\
&\quad + \backslash permittivity_{ik} \backslash efield_k \\
&\quad + \backslash pyroelec_i \backslash temperature \\
\backslash Delta \backslash entropydensity &= \backslash thermalexpansion_{kl} \backslash stress_{kl} \\
&\quad + \backslash pyroelec_k \backslash efield_k \\
&\quad + (\backslash heatcapacity / \backslash temperature) \\
&\quad \backslash Delta \backslash temperature.
\end{align*}
}

```

for a set of linear constitutive equations. It renders as

$$\begin{aligned}
\epsilon_{ij} &= s_{ijkl}^{E,T} \sigma_{kl} + d_{kij}^T E_k + \alpha_{ij}^E \Delta T \\
D_i &= d_{ikl}^T \sigma_{kl} + \kappa_{ik}^{\sigma,T} E_k + p_i^\sigma \Delta T \\
\Delta s &= \alpha_{kl}^E \sigma_{kl} + p_k^\sigma E_k + (C^{\sigma,E}/T) \Delta T.
\end{aligned}$$

### 3 Defining notation

## A Thermodynamic potentials and equations of state

In this section, we rely on Mason’s description in [?] and stick to his notation. The constitutive equations can be derived from the thermodynamic potentials in table ??.

Table 3: Definitions of thermodynamic potential (densities) according to Mason [?].

| Name                    | Symbol | Free variables   | Definition                  | Differential                            |
|-------------------------|--------|------------------|-----------------------------|-----------------------------------------|
| Internal energy         | $U$    | $S_{ij}, D_k, S$ | –                           | $dU = T_{ij}dS_{ij} + E_kdD_k + TdS$    |
| Enthalpy                | $H$    | $T_{ij}, E_k, S$ | $U - T_{ij}S_{ij} - E_kD_k$ | $dH = -S_{ij}dT_{ij} - D_kdE_k + TdS$   |
| Elastic enthalpy        | $H_1$  | $T_{ij}, D_k, S$ | $U - T_{ij}S_{ij}$          | $dH_1 = -S_{ij}dT_{ij} + E_kdD_k + TdS$ |
| Electric enthalpy       | $H_2$  | $S_{ij}, E_k, S$ | $U - E_kD_k$                | $dH_2 = T_{ij}dS_{ij} - D_kdE_k + TdS$  |
| Free energy             | $A$    | $S_{ij}, D_k, T$ | $U - TS$                    | $dA = T_{ij}dS_{ij} + E_kdD_k - SdT$    |
| Gibbs function          | $G$    | $T_{ij}, E_k, T$ | $A - T_{ij}S_{ij} - E_kD_k$ | $dG = -S_{ij}dT_{ij} - D_kdE_k - SdT$   |
| Elastic Gibbs function  | $G_1$  | $T_{ij}, D_k, T$ | $A - T_{ij}S_{ij}$          | $dG_1 = -S_{ij}dT_{ij} + E_kdD_k - SdT$ |
| Electric Gibbs function | $G_2$  | $S_{ij}, E_k, T$ | $A - E_kD_k$                | $dG_2 = T_{ij}dS_{ij} - D_kdE_k - SdT$  |

From the table, we see that the fields can be given in terms of derivatives of the thermodynamic potentials as

$$T_{ij} = \frac{\partial U}{\partial S_{ij}} = \frac{\partial H_2}{\partial S_{ij}} = \frac{\partial A}{\partial S_{ij}} = \frac{\partial G_2}{\partial S_{ij}}, \quad (1)$$

$$E_k = \frac{\partial U}{\partial D_k} = \frac{\partial H_1}{\partial D_k} = \frac{\partial A}{\partial D_k} = \frac{\partial G_1}{\partial D_k}, \quad (2)$$

$$T = \frac{\partial U}{\partial S} = \frac{\partial H}{\partial S} = \frac{\partial H_1}{\partial S} = \frac{\partial H_2}{\partial S}, \quad (3)$$

$$S_{ij} = -\frac{\partial H}{\partial T_{ij}} = -\frac{\partial H_1}{\partial T_{ij}} = -\frac{\partial G}{\partial T_{ij}} = -\frac{\partial G_1}{\partial T_{ij}}, \quad (4)$$

$$D_k = -\frac{\partial H}{\partial E_k} = -\frac{\partial H_2}{\partial E_k} = -\frac{\partial G}{\partial E_k} = -\frac{\partial G_2}{\partial E_k}, \quad (5)$$

$$S = -\frac{\partial U}{\partial T} = -\frac{\partial H}{\partial T} = -\frac{\partial H_1}{\partial T} = -\frac{\partial H_2}{\partial T}. \quad (6)$$

The linear coefficients can then be defined as the first derivatives of a field under certain conditions on the other free variables or as a second derivative of a potential. For stiffness and compliance, we have

$$c_{ijkl}^{D,S} = \frac{\partial^2 U}{\partial S_{kl} \partial S_{ij}}, \quad c_{ijkl}^{E,S} = \frac{\partial^2 H_2}{\partial S_{kl} \partial S_{ij}}, \quad c_{ijkl}^{D,T} = \frac{\partial^2 A}{\partial S_{kl} \partial S_{ij}}, \quad c_{ijkl}^{E,T} = \frac{\partial^2 G_2}{\partial S_{kl} \partial S_{ij}}, \quad (7)$$

$$s_{ijkl}^{E,S} = -\frac{\partial^2 H}{\partial T_{kl} \partial T_{ij}}, \quad s_{ijkl}^{D,S} = -\frac{\partial^2 H_1}{\partial T_{kl} \partial T_{ij}}, \quad s_{ijkl}^{E,T} = -\frac{\partial^2 G}{\partial T_{kl} \partial T_{ij}}, \quad s_{ijkl}^{D,T} = -\frac{\partial^2 G_1}{\partial T_{kl} \partial T_{ij}}. \quad (8)$$

## References

- [1] Publication and proposed revision of ANSI/IEEE standard 176-1987 "ANSI/IEEE standard on piezoelectricity". *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 43(5):717–, Sept 1996.
- [2] Takuro Ikeda. *Fundamentals of Piezoelectricity*. Oxford University Press Inc., New York, 1996.
- [3] Warren P. Mason. *Physical acoustics and the properties of solids*. D. Van Nostrand Company, Inc, Princeton, 1958.
- [4] J. F. Nye. *Physical Properties of Crystals*. Oxford University Press, Oxford, 1985.